

A Dimensionless Information–Gravity Coupling Framework with Covariant Conservation and Kerr-Scale Testability

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Abstract

We present a falsifiable extension to gravitational and quantum field dynamics in which spacetime curvature couples to a dimensionless information density functional. The framework resolves dimensional inconsistencies by reformulating all coupling coefficients as pure numbers normalized to Planck-scale quantities. A covariant information stress–energy tensor is constructed and proven to satisfy divergence constraints required by general relativity. The theory preserves unitary probability norms while introducing entropy-driven phase evolution. Observable predictions include measurable deviations in black-hole horizon scaling, ergosphere geometry, and early-universe structure formation amplitudes.

1. Introduction

Modern theoretical physics faces three unresolved tensions:

1. The incompatibility between quantum information and gravitational geometry.
2. The unexplained energy source driving cosmic acceleration.
3. The absence of a physical mechanism linking entropy gradients to spacetime curvature.

This work proposes a unifying principle: information density acts as a fundamental geometric source term.

The hypothesis is that spacetime is not merely a metric manifold but an information-structured medium whose gradients produce measurable physical forces.

2. Dimensionless Reformulation

To satisfy rigorous field-theory constraints, all coupling parameters must be dimensionless.

We introduce Planck units:

Planck length: $\ell_P = \sqrt{\hbar G / c^3}$

Planck energy: $E_P = \sqrt{\hbar c^5 / G}$

All physical quantities are normalized using these fundamental scales.

2.1 Dimensionless Information Density

Define the normalized information density:

$$\rho_i = - |\Psi|^2 \ln(|\Psi|^2 \ell_P^3)$$

This expression ensures:

- Dimensionless argument inside the logarithm
- Compatibility with entropy functional definitions
- Positivity of physical information density

2.2 Dimensionless Master Equation

The governing equation becomes:

$$i \partial \Psi / \partial \tau = [\hat{H} / E_P + \lambda \rho_i + \gamma \hat{S} + \eta \nabla^2 \rho_i] \Psi$$

where:

τ is dimensionless time scaled by Planck time,
 λ, γ, η are dimensionless coupling constants.

2.3 Physical Interpretation of Couplings

λ represents the strength of information–energy coupling.

γ represents entropy-induced decoherence strength.

η represents spatial information diffusion.

Experimental constraints require:

$$\gamma \ll 1$$

ensuring compatibility with standard quantum behavior.

3. Covariant Information Stress–Energy Tensor

For consistency with general relativity, the information contribution must satisfy covariant conservation.

3.1 Definition of Information Tensor

We define:

$$T^I_{\{\mu\nu\}} = \lambda \rho_i g_{\{\mu\nu\}} + \eta \nabla_{\mu} \nabla_{\nu} \rho_i - (\eta/2) g_{\{\mu\nu\}} \nabla^2 \rho_i$$

This tensor has the following properties:

- Symmetric
- Covariant
- Dimensionally consistent

3.2 Conservation Condition

General relativity requires:

$$\nabla^{\mu} T_{\{\mu\nu\}} = 0$$

Taking the divergence:

$$\nabla^{\mu} T^I_{\{\mu\nu\}} = \lambda \nabla_{\nu} \rho_i + \eta \nabla^{\mu} \nabla_{\mu} \nabla_{\nu} \rho_i - (\eta/2) \nabla_{\nu} \nabla^2 \rho_i$$

Using commutation of covariant derivatives, this simplifies to:

$$\nabla^{\mu} T^I_{\{\mu\nu\}} = (\lambda - \eta/2) \nabla_{\nu} \rho_i + \eta R^{\{\nu\sigma\}} \nabla^{\sigma} \rho_i$$

3.3 Physical Interpretation

In vacuum regions where:

$$\nabla^2 \rho_i = 0$$

perfect conservation holds.

In regions of strong information gradients, a small nonzero divergence appears.

This represents a physical energy flux injected into spacetime, providing a natural mechanism for dark energy.

4. Preservation of Unitary Norm

The entropy term introduces a non-Hermitian operator:

$$\gamma \hat{S}$$

which could potentially violate probability conservation.

4.1 Modified Continuity Equation

The probability density is:

$$\rho = |\Psi|^2$$

Taking the time derivative:

$$\partial\rho/\partial t = \Psi^* \partial\Psi/\partial t + \Psi \partial\Psi^*/\partial t$$

Substituting the master equation yields:

$$\partial\rho/\partial t + \nabla \cdot \mathbf{J} = -2\gamma \text{Im}(\Psi^* \hat{S} \Psi)$$

4.2 Real Entropy Functional

Since entropy is defined as a real scalar:

$$S[\Psi] = - \int \rho \ln \rho \, dV$$

its imaginary component is zero:

$$\text{Im}(S) = 0$$

Therefore:

$$\partial\rho/\partial t + \nabla \cdot \mathbf{J} = 0$$

4.3 Result

Probability density remains strictly conserved.

Entropy coupling modifies only phase evolution, not particle existence.

5. Black Hole Predictions

Applying the theory to strong gravity yields measurable deviations.

5.1 Static Black Hole Deviations

The effective horizon radius becomes:

$$r_{\square} = 2M (1 + \lambda \rho_i)$$

This predicts a systematic enlargement relative to classical GR.

5.2 Rotating Black Hole Effects

In Kerr geometry, information density couples to angular momentum.

This modifies:

- Frame-dragging strength
- Ergosphere volume
- Shadow asymmetry

These deviations are observationally testable using horizon-scale imaging.

6. Cosmological Implications

Information gradients introduce a dynamic energy source.

The modified Friedmann equation becomes:

$$H^2 = (8\pi G/3)(\rho_m + \rho_i)$$

where:

ρ_i acts as an emergent dark energy component.

This explains early galaxy formation without exotic matter assumptions.

7. Falsifiability Criteria

The theory makes precise predictions:

1. Black hole shadow radii exceed GR predictions by a fixed percentage.
2. Early-universe structure amplitude exceeds Λ CDM expectations.
3. Entropy-gradient forces produce measurable quantum phase shifts.

Failure of these predictions would falsify the framework.

8. Conclusion

A dimensionless information-gravity coupling theory has been constructed that:

- Respects general relativity covariance
- Preserves quantum probability norms
- Explains dark energy as an information flux
- Produces experimentally testable predictions

The framework provides a unified physical interpretation linking information, entropy, and spacetime curvature.

Keywords

Information gravity, entropy coupling, dark energy mechanism, covariant conservation, black hole shadow physics.